

Descriptive Written test

Smart Home Automation System

→ A smart home automation system controls lighting, temp., and security using information from sensors such as motion detectors, temperature sensors, cameras, and clocks.

Types of intelligent Agents

a) simple reflex agents

Agent which makes decisions based only on the current input without considering past events.

Examples in Smart Home:

- If motion is detected → Turn on the lights.
- If temp. is above 30°C → Turn on the air conditioner.
- If no motion is detected for 10 minutes → Turn off the lights.

Advantages

Fast response

Easy to implement

Suitable for simple tasks.

Disadvantages:

- Cannot remember previous states
- Cannot learn user preferences.
- May make incorrect decisions in complex situations.

b) Model-Based

Maintains an internal model of the environment and remembers previous states.

Example in smart homes:

- Knows that the user is sleeping at night.
- keeps track of room occupancy.

Advantages:

- Improves reliability
- Makes better decisions using past information.

Disadvantages:

- More memory and computation required.
- More complex than a simple reflex agent.

c) Goal-based agent

A Goal-based agent chooses actions that help achieve specific goals.

Ex:- Maintains room temp. at 24°C .

Reduced electricity consumption.

Advantages:-

Makes intelligent decisions based on goals.

Can evaluate different possible actions.

Disadvantages:-

Requires planning

May take more time to make decisions.

d) Utility - Based Agent:

A utility-based agent selects the action that provides highest overall benefit (utility) by considering multiple factors.

Ex:- The system balances:

User comfort

Energy consumption

Home security

Electricity cost

Advantages:-

Produces the best overall decision.

Handles trade-offs effectively.

Disadvantages:-

Most complex to design.

Requires utility calculations and learning mechanisms.

Conclusion

A hybrid system using model-based and utility-based systems agent provides the best balance between comfort, security and energy efficiency.

2) Robotic in an unknown maze - UCS & IDS.

A robot is placed in an unknown maze and must find the exit without knowing the map. It explores different paths until it reaches the goal. Search algorithms help the robot decide which path to follow.

UCS (Uniform Cost Search)

Working,

Always expands the node with the lowest cumulative path cost.

Ex:- If one path costs 5 units and another costs 8 units, UCS explores the path with cost 5 units first.

Advantages

- Guarantees the lowest-cost path.
- Complete if path costs are positive.

Disadvantages:

- High memory usage.
- Slow for large search spaces.
- Stores many nodes.

Complexity:

Time Complexity: Exponential (depends on branching factor and path cost).

Space Complexity: Exponential.

Iterative deepening Search (IDS)

Working

IDS performs repeated depth-first search with increasing depth limits until the goal is found.

Ex:-

Depth 0

Depth 1

Depth 2

- Continue until the exit is discovered.

Advantages

- Uses very little memory
- Finds shallow solutions efficiently

Disadvantages

- Repeats node expansions
- Not suitable when path costs differ.

Complexity:-

Time complexity: $O(b^d)$

Space complexity: $O(b \times d)$

where;

b = branching factor

d = depth of the goal.

Conclusion

A goal-based ~~performed~~ agent combined with uniform cost search provides optimal navigation by finding the least-cost path, while ~~iterative~~ deepening search when memory is limited.